

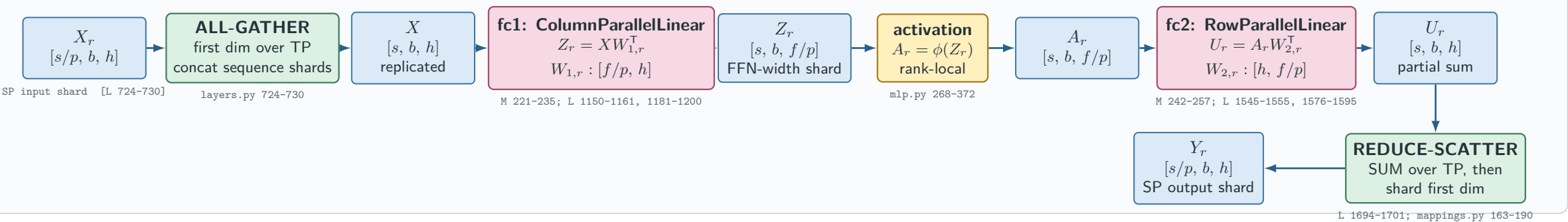
Megatron-Core tensor + sequence parallel MLP on TP rank r

one rank shown: $r \in \{0, \dots, p-1\}$

Global sequence s is evenly split over p TP ranks; tensor order is [sequence, b , channel]; f = ffn_hidden_size.

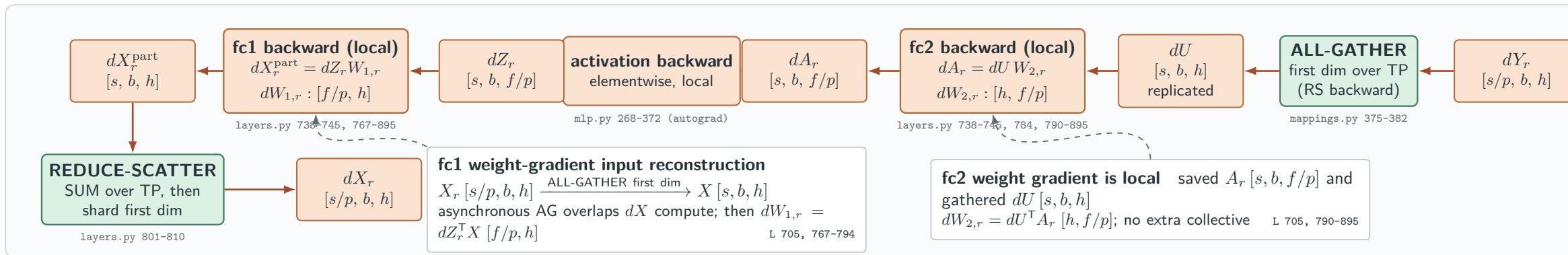
FORWARD

column-parallel fc1 $\rightarrow$ local activation $\rightarrow$ row-parallel fc2



BACKWARD

autograd reverses fc2's RS to AG; fc1 directly launches RS for dX



Scope / variants. Main path is the trainable, dense, non-gated MLP (gated_linear_unit=False), with s and f divisible by p . For GLU/SwiGLU, fc1 and its pre-activation use $2f/p$; the gated activation returns f/p before fc2. mlp.py 203-215, 330-349.

Bias paths (not drawn on the main arrows). fc1 bias is sharded $b_{1,r} : [f/p]$ and is added/fused with the activation; fc2 bias is replicated $b_2 : [h]$ and returned separately because both layers use skip_bias_add=True. mlp.py 226-251, 268-328; layers.py 1222-1240, 1619-1636, 1704-1710.

Weight orientation. Shapes shown are the actual stored PyTorch parameters: F.linear evaluates XW^T . fc1 shards stored axis 0 (output features); fc2 shards stored axis 1 (input features). layers.py 1150-1200, 1545-1595.

Pinned code key M=megatron/core/transformer/mlp.py; L=megatron/core/tensor_parallel/layers.py. Collective primitives and autograd duals: mappings.py 122-160 (AG first dim), 163-202 (RS first dim), 304-356 (AG backward $\rightarrow$ RS), 359-385 (RS backward $\rightarrow$ AG), 528-549 (public wrappers). Commit e998be072d22ff57b4b49899819a1a153f4f3dc9.